

Online monotone density estimation and log-optimal calibration

Rohan Hore

RHORE@ANDREW.CMU.EDU

*Department of Statistics and Data Science
Carnegie Mellon University
USA*

Ruodu Wang

WANG@UWATERLOO.CA

*Department of Statistics and Actuarial Science
University of Waterloo
Canada*

Aaditya Ramdas

ARAMDAS@STAT.CMU.EDU

*Department of Statistics and Data Science
Carnegie Mellon University
USA*

Editor: Ernst Ahlberg, Ulf Johansson, Henrik Boström, Alberto Carlevaro, Johan Hallberg Szabadváry and Lars Carlsson

Abstract

We study the problem of online monotone density estimation, where density estimators must be constructed in a predictable manner from sequentially observed data. We propose two online estimators: an online analogue of the classical Grenander estimator, and an expert aggregation estimator inspired by exponential weighting methods from the online learning literature. In the well-specified stochastic setting, where the underlying density is monotone, we show that the expected cumulative log-likelihood gap between the online estimators and the true density admits an $O(n^{1/3})$ bound. We further establish a $\sqrt{n \log n}$ pathwise regret bound for the expert aggregation estimator relative to the best offline monotone estimator chosen in hindsight, under minimal regularity assumptions on the observed sequence. As an application of independent interest, we show that the problem of constructing log-optimal p-to-e calibrators for sequential hypothesis testing can be formulated as an online monotone density estimation problem. We adapt the proposed estimators to build empirically adaptive p-to-e calibrators and establish their optimality. Numerical experiments illustrate the theoretical results.

Keywords: Online monotone density estimation, online Grenander estimator, expert aggregation, p-to-e calibration

1. Introduction

In this work, we study the problem of online monotone density estimation. Let $\mathcal{D}$ be the class of non-increasing densities supported on $[0, 1]$, defined by

$$\mathcal{D} := \left\{ f : [0, 1] \rightarrow [0, \infty) : \int f(u) du = 1, f(x) \leq f(y) \text{ for all } x \geq y \right\}. \quad (1)$$

Suppose $X_1, X_2, \dots \stackrel{iid}{\sim} \mathbb{Q}$ are observations drawn sequentially, where $\mathbb{Q}$ admits a density $q = \frac{d\mathbb{Q}}{d\lambda} \in \mathcal{D}$. Throughout this paper, λ is the Lebesgue measure on $[0, 1]$. Our aim is to

construct an *algorithm* $\hat{f}$, by which we mean a predictable sequence of density estimators $\hat{f} = (\hat{f}_t)_{t \geq 1}$. Formally, at each time $t \geq 2$, the algorithm outputs a density $\hat{f}_t$, a measurable function of the past observations $(X_1, \dots, X_{t-1})$, that is, $\hat{f}_t$ is $\mathcal{F}(X_1, \dots, X_{t-1})$ -measurable, where $\mathcal{F}(X_1, \dots, X_{t-1})$ denotes the σ -algebra generated by $(X_1, \dots, X_{t-1})$.

Restricting attention to densities supported on a compact interval is standard in monotone density estimation and is often essential for meaningful risk bounds; see, for example, (Gao and Wellner, 2009, Section 1). While in this paper, we assume densities are supported on $[0, 1]$, any compact interval can be studied by suitable translation and rescaling.

1.1. Risk and regret for online monotone density estimation

We measure the performance of an online algorithm $\hat{f}$ through its sequential log-likelihood. In particular, for a predictable density $f \in \mathcal{D}$, we define $\ell_t(f) := -\log f(X_t)$, the negative log-likelihood of the observation X_t , as the instantaneous loss at time t . Accordingly, the cumulative loss incurred by an online algorithm $\hat{f}$ up to time n is then given by

$$\mathcal{L}(\hat{f}, n) = \sum_{t=1}^n \ell_t(\hat{f}_t) = -\sum_{t=1}^n \log \hat{f}_t(X_t).$$

For any fixed f , we will write $\mathcal{L}(f, n)$ to denote $-\sum_{t=1}^n \log f(X_t)$. We study this loss under both stochastic and adversarial regimes, comparing the online algorithm with the population-level and hindsight-optimal benchmarks.

Stochastic well-specified setting. When $X_1, X_2, \dots$ are iid observations from $\mathbb{Q}$, with density $q \in \mathcal{D}$, the true density q is Bayes-optimal under log-loss. This motivates the following notion of *excess KL-risk*,

$$\text{Risk}_{\mathbb{Q}}(\hat{f}; n) = \mathbb{E}_{\mathbb{Q}}[\mathcal{L}(\hat{f}, n)] - \min_{f \in \mathcal{D}} \mathbb{E}_{\mathbb{Q}}[\mathcal{L}(f, n)] = \mathbb{E}_{\mathbb{Q}}[\mathcal{L}(\hat{f}, n)] - \mathbb{E}_{\mathbb{Q}}[\mathcal{L}(q, n)]. \quad (2)$$

For any online algorithm $\hat{f}$, since $\hat{f}_s$ is $\mathcal{F}_{s-1}$ -measurable, by the tower property, we can express excess KL-risk equivalently as $\sum_{s=1}^n \mathbb{E}[\text{KL}(q \| \hat{f}_s)]$.

Adversarial setting. We also compare an online algorithm against the best monotone density that could be selected offline with full access to the data up to a fixed time $n \geq 1$, in order to assess how well the online procedures track a hindsight-optimal benchmark. For $0 \leq a \leq b \leq \infty$, let

$$\mathcal{D}_{a,b} := \{f \in \mathcal{D} : a \leq f(u) \leq b \text{ for all } u \in [0, 1]\}$$

denote the subclass of $\mathcal{D}$ consisting of densities that are uniformly bounded below and above by a and b , respectively. A natural offline benchmark is given by the solution of the maximum likelihood problem constrained to $\mathcal{D}_{a,b}$,

$$\tilde{f}_{n,a,b}^{\text{MLE}} \in \operatorname{argmax}_{f \in \mathcal{D}_{a,b}} \sum_{i=1}^n \log f(X_i). \quad (3)$$

The unconstrained case $a = 0$ and $b = \infty$ was originally introduced by Grenander (1956) and is known as the Grenander density estimator in the literature; therefore, we refer to (3) as

the *constrained Grenander estimator*. Comparing an online algorithm $\hat{f}$ to this benchmark leads to:

$$\text{Regret}(\hat{f}; n, a, b) := \mathcal{L}(\hat{f}, n) - \min_{f \in \mathcal{D}_{a,b}} \mathcal{L}(f, n), \quad (4)$$

which provides a pathwise performance measure without imposing any distributional assumptions.

For theoretical analysis, we will often restrict attention to the bounded class $\mathcal{D}_{a,b}$. In practice, however, one may take $a = 0$ and $b = \infty$, recovering the classical Grenander estimator. Ideally, a well-designed online procedure should adapt automatically to the effective range of the density, without requiring prior knowledge of such bounds. Empirically, we indeed observe this behavior for all online procedures developed in this work.

1.2. The Grenander estimator and its properties

[Grenander \(1956\)](#) characterized the Grenander estimator as the left-continuous density corresponding to the least concave majorant of the empirical distribution function. Consequently, the Grenander estimator is simply a monotone histogram that is strictly positive on $[0, \max_{1 \leq i \leq n} X_i]$. Throughout this work, we extend the estimator beyond $\max_{1 \leq i \leq n} X_i$ by a right-continuous constant equal to the final histogram height and suitably renormalize it, thereby ensuring positivity and finite log-loss evaluations. Since the foundational work of [Grenander \(1956\)](#), the theoretical properties of the Grenander estimator have been studied extensively; see [Durot et al. \(2012\)](#) for a comprehensive historical overview.

Early work by [Rao \(1969\)](#) and [Carolán and Dykstra \(1999\)](#) characterized the asymptotic distribution of $\hat{f}_n^{\text{MLE}}(x)$ in regions where the true density is locally strictly decreasing and locally flat, respectively. Later studies established sharp global convergence rates under various loss functions. In particular, convergence at rate $n^{-1/3}$ in Hellinger distance was proved in [Van de Geer \(1993\)](#); [Gao and Wellner \(2009\)](#), while asymptotic normality of the L_1 error was analyzed in [Groeneboom et al. \(1999\)](#). Additional asymptotic and non-asymptotic properties of the Grenander estimator have been developed in [Balabdaoui and Wellner \(2010\)](#); [Birge \(1989\)](#); [Kulikov and Lopuhaä \(2005\)](#); [Groeneboom and Lopuhaä \(1993\)](#).

1.3. Our contributions

Online monotone density estimation. Despite the extensive offline literature, monotone density estimation has not been systematically studied in the online setting. This work fills this gap.

- We propose two online monotone density estimators: an online analogue of the classical Grenander estimator, and an expert aggregation estimator based on exponential weighting methods from online learning.
- In the well-specified stochastic setting, where the true density q is monotone, uniformly bounded above and bounded away from zero, we establish statistical guarantees for both estimators, showing that their excess KL-risk is of order $n^{1/3}$.

- Next, we study a semi-adversarial regime and derive a pathwise regret bound of order $\sqrt{n \log n}$ for the expert aggregation estimator, under mild regularity conditions on the observed sequence.
- The pathwise regret bound suggests improved finite-sample adaptivity of the expert aggregation estimator relative to the online Grenander estimator in a mis-specified stochastic setting, a phenomenon that we confirm empirically through numerical simulations.

Application to sequential testing. We develop a non-trivial application of online monotone density estimation to the construction of sequential hypothesis tests.

- While most classical tests are formulated in terms of p-values, e-values provide a flexible framework for sequential inference under optional stopping. We observe that the problem of constructing optimal p-to-e calibrators reduces to online monotone density estimation.
- We adapt the proposed online density estimators to develop empirically adaptive p-to-e calibrators, yielding valid sequential tests. If under the alternative, the p-values are iid with a monotone density, the resulting calibrators are asymptotically log-optimal.

Organization of the paper. Section 2 introduces the proposed online monotone density estimators and establishes bounds on their excess KL-risk and regret. Section 3 presents an application to sequential testing, yielding valid procedures with provable optimality guarantees. Finally, Section 4 illustrates the theoretical results through a series of numerical experiments.

2. Online monotone density estimation

In this section, we introduce two online algorithms and then establish their theoretical properties.

Online Grenander algorithm. Motivated by the classical Grenander estimator $\tilde{f}_n^{\text{MLE}}$, a natural approach is to adapt the maximum likelihood formulation to the online setting. Specifically, for any $0 \leq a \leq b < \infty$, the online Grenander (OG) algorithm $\hat{f}_{a,b}^{\text{OG}} = (\hat{f}_{t,a,b}^{\text{OG}})_{t \geq 1}$ produces, at each time $t \geq 2$, the estimator

$$\hat{f}_{t,a,b}^{\text{OG}} \in \operatorname{argmax}_{f \in \mathcal{D}_{a,b}} \sum_{i=1}^{t-1} \log f(X_i), \quad \hat{f}_{1,a,b}^{\text{OG}} \equiv 1. \quad (5)$$

This algorithm is the direct online analogue of the constrained offline MLE $\tilde{f}_{n,a,b}^{\text{MLE}}$. While statistically natural, it recomputes a full monotone MLE at each time step.

Expert aggregation algorithm. To address the computational inefficiency of OG, we consider an aggregation-based alternative, well studied in the online learning literature (Cesa-Bianchi and Lugosi, 2006; Hao et al., 2018). Fix any $0 \leq a \leq b < \infty$ and a finite collection of monotone density experts

$$\mathcal{E}_m = \{f_1, \dots, f_m\} \subseteq \mathcal{D}_{a,b}.$$

The expert aggregation (EA) algorithm $\hat{f}_{a,b}^{\text{EA}} = (\hat{f}_{t,a,b}^{\text{EA}})_{t \geq 1}$ outputs, at each time $t \geq 1$, the estimator

$$\hat{f}_{t,a,b}^{\text{EA}}(u) = \sum_{f \in \mathcal{E}_m} w_t(f) f(u), \quad w_t(f) = \frac{\prod_{i=1}^{t-1} f(X_i)}{\sum_{f' \in \mathcal{E}_m} \prod_{i=1}^{t-1} f'(X_i)}, \quad u \in [0, 1]. \quad (6)$$

Thus, the EA algorithm produces a predictable sequence of estimators formed as convex combinations of experts, with weights favoring densities that assign higher likelihood to past observations.

2.1. Risk bounds for online monotone density estimation

We now bound the excess KL-risk, defined in (2), for both the OG algorithm (5) and the EA algorithm (6), which serves as a measure of the statistical accuracy of the proposed online procedures in estimating the true monotone density q . Throughout this section, we assume that

$$X_1, X_2, \dots \stackrel{iid}{\sim} \mathbb{Q}, \quad q := \frac{d\mathbb{Q}}{d\lambda} \in \mathcal{D}.$$

To place our results in context, we first recall some relevant theoretical results of the Grenander estimator. If q is bounded above, then the Grenander estimator satisfies $d_H(q, \tilde{f}_n^{\text{MLE}}) = O_p(n^{-1/3})$ (Gao and Wellner, 2009; Van de Geer, 1993), where d_H denotes the Hellinger distance. When both q and the estimator are additionally bounded away from zero and infinity, this result implies a corresponding $n^{-2/3}$ rate for the expected Kullback–Leibler divergence. We state the precise implication for the KL risk formally below, for later use in our analysis.

Lemma 1 *Fix $n \in \mathbb{N}$. Suppose that there exist $0 < a < b < \infty$ such that $q \in \mathcal{D}_{a,b}$. Then, the constrained Grenander estimator $\hat{f}_{n,a,b}^{\text{MLE}} \equiv \hat{f}_{n,a,b}^{\text{MLE}}(X_1, \dots, X_n)$ satisfies*

$$\mathbb{E}_{\mathbb{Q}}[\text{KL}(q \parallel \hat{f}_{n,a,b}^{\text{MLE}})] \leq \kappa n^{-2/3}$$

for some constant $\kappa > 0$ depending only on a and b .

We are now ready to state the bounds on excess KL-risk for OG and the EA estimators.

Theorem 2 (Risk bounds for OG and EA algorithms) *Fix $n \in \mathbb{N}$ and $0 < a < b < \infty$. For any $q \in \mathcal{D}_{a,b}$ and $X_1, \dots, X_n \stackrel{iid}{\sim} \mathbb{Q}$, the OG algorithm satisfies*

$$\text{Risk}_{\mathbb{Q}}(\hat{f}_{a,b}^{\text{OG}}; n) \leq \Gamma_{\text{OG}}(a, b) n^{1/3},$$

and there exists a collection of monotone density experts $\mathcal{E}_{m_n}$ with size $m_n < \infty$ such that for all $q \in \mathcal{D}_{a,b}$, the corresponding EA algorithm satisfies

$$\text{Risk}_{\mathbb{Q}}(\hat{f}_{a,b}^{\text{EA}}; n) \leq \Gamma_{\text{EA}}(a, b) n^{1/3},$$

where the constants $\Gamma_{\text{OG}}(a, b)$ and $\Gamma_{\text{EA}}(a, b)$ depend only on a and b .

The rate in Theorem 2 is essentially optimal. Indeed, in the offline setting, monotone density estimation is known to attain minimax L_1 risk of order $n^{-1/3}$, corresponding to excess-KL risk of order $n^{1/3}$. Moreover, any substantial improvement in the online rate would yield an improvement in the offline problem through a standard online-to-offline conversion argument. Specifically, given online estimators $\hat{f}_1, \dots, \hat{f}_n$, one may construct an offline estimator via averaging,

$$\bar{f}_n := \frac{1}{n} \sum_{t=1}^n \hat{f}_t,$$

thereby transferring cumulative online risk guarantees to the offline setting.

Although both procedures achieve the same worst-case rate, their finite-sample behavior can differ substantially. In particular, our empirical results suggest that the EA procedure often attains smaller regret in practice, reflecting improved empirical adaptivity.

2.2. Bounding regret for the EA algorithm

In this subsection, we fix $n \in \mathbb{N}$, $0 < a < b < \infty$ and derive a pathwise regret bound for the EA algorithm under mild spacing and boundary conditions on the observed sequence $(x_1, x_2, \dots, x_n)$. The proof is based on a carefully chosen finite class of monotone histogram experts, that we define below.

We begin by introducing a binning of the unit interval. For $\beta > 0$ and $n \geq 1$, define the grid

$$\mathcal{B}_n := \left\{0, \Delta_b, 2\Delta_b, \dots, \lfloor n^{(\beta+1)} \rfloor \Delta_b\right\}, \quad \Delta_b := n^{-(\beta+1)}.$$

Building upon this grid, we construct a finite class of monotone histogram densities. For integers $k \geq 1$, let $\mathcal{E}_{k,n,\beta}$ denote the collection of functions $g \in \mathcal{D}_{a,b}$ of the form

$$g(x) = \sum_{j=1}^r \theta_j \mathbb{1}_{\{t_{j-1} \leq x < t_j\}},$$

where the parameters satisfy the following:

- (i) the number of bins r satisfies $r \leq k$;
- (ii) the breakpoints $t_1, \dots, t_{r-1}$ belong to the grid $\mathcal{B}_n$, and $t_0 = 0, t_r = 1$;
- (iii) the heights are non-increasing, $\theta_1 \geq \dots \geq \theta_{r-1}$, with $\log \theta_j \in \Lambda_{n,a,b}$ for $j = 1, \dots, r-1$, where $\Lambda_{n,a,b} := \log(a) + \log(b/a) \mathcal{B}_n$;
- (iv) when $r \geq 2$, the final height θ_r is defined as $\theta_r := \frac{1 - \sum_{j=1}^{r-1} \theta_j (t_j - t_{j-1})}{1 - t_{r-1}}$, and is required to satisfy that $\theta_r \in [a, b]$ and $\theta_{r-1} \geq \theta_r$.

If $r = 1$, then only feasible function is $g \equiv 1$. We show in the next section that, for fixed (n, k, β) , we derive a bound on the size of $\mathcal{E}_{k,n,\beta}$. Classical discretizations of monotone density classes, as in Gao and Wellner (2007), are typically designed to control approximation error in metrics such as L_p or Hellinger distance. Such guarantees are insufficient for the present analysis, which requires control of cumulative log-loss and hence control on pointwise likelihood ratios along the realized sample path. Our construction therefore refines these classical discretization ideas to additionally provide suitable pointwise control while preserving monotonicity.

The expert class $\mathcal{E}_{k,n,\beta}$ is introduced solely to establish the pathwise regret guarantee, stated below. It differs from the class used in the proof of Theorem 2, which is tailored to control excess risk. In particular, $\mathcal{E}_{k,n,\beta}$ depends on the finite horizon n through the break-point and height discretization choices. This growing, horizon-dependent class is primarily a theoretical device; in practice, one may aggregate over a much smaller fixed collection of experts and still efficiently estimate the underlying density, as demonstrated in Section 4.

Before stating the regret bound, we formalize the regularity conditions imposed on the data sequence. For $\beta, \gamma > 0$ and $n \geq 1$, we define the *good* set

$$\mathcal{S}_n(\beta, \gamma) := \left\{ (x_1, \dots, x_n) \in [0, 1]^n : \min_{1 \leq i < j \leq n} |x_i - x_j| \geq n^{-\beta}, \max_{1 \leq i \leq n} x_i \leq 1 - n^{-\gamma} \right\}. \quad (7)$$

Elements of $\mathcal{S}_n(\beta, \gamma)$ are well separated, with minimum spacing of order $n^{-\beta}$, and remain uniformly away from the boundary point 1 by a margin of order $n^{-\gamma}$. These conditions ensure stability of the discretization with respect to the pathwise loss along the observed sample path from $\mathcal{S}_n(\beta, \gamma)$. In particular, the rounding of breakpoints and heights induced by the experts in $\mathcal{E}_{k,n,\beta}$ introduces only a controlled perturbation in likelihood evaluations, effectively yielding local ℓ_∞ control along the sample path. We are now ready to formally state the regret bound.

Theorem 3 *Fix $\beta > 1/2$, $\gamma \in (0, \beta - \frac{1}{2})$. For all sufficiently large n , and for all $(x_1, \dots, x_n) \in \mathcal{S}_n(\beta, \gamma)$, the EA algorithm $\hat{f}_{a,b}^{\text{EA}}$ based on the set of experts $\mathcal{E}_{k,n,\beta}$ with $k = \lfloor (n/\log n)^{1/2} \rfloor$ satisfies*

$$\text{Regret}(\hat{f}_{a,b}^{\text{EA}}; n, a, b) \leq \Gamma(a, b, \beta) \sqrt{n \log n},$$

where $\Gamma(a, b, \beta)$ is a constant depending only on a , b , and β .

Although the regularity conditions encoded in $\mathcal{S}_n(\beta, \gamma)$ may appear restrictive, they are mild in the stochastic setting. In particular, when $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathbb{Q}$ with $q \in \mathcal{D}$, for any choices of $\beta > 2$ and any $\gamma > 1$, the sample path $(X_1, \dots, X_n)$ belongs to $\mathcal{S}_n(\beta, \gamma)$ with high probability. We formalize these statements in Lemma 11, given in Appendix B. As an immediate consequence, we get the following high-probability regret bound under the iid alternative model.

Corollary 4 *Suppose $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathbb{Q}$ with $q \in \mathcal{D}_{a,b}$. Consider the EA algorithm $\hat{f}_{a,b}^{\text{EA}}$ based on the set of experts $\mathcal{E}_{k,n,\beta}$ with $k = \lfloor (n/\log n)^{1/2} \rfloor$ for any $\beta > 2$. Then, for any $\delta \in (0, 1)$, there exists $N \equiv N(\delta, \beta, b)$ such that for all $n > N$,*

$$\mathbb{P}_{\mathbb{Q}} \left(\text{Regret}(\hat{f}_{a,b}^{\text{EA}}; n, a, b) \leq C \sqrt{n \log n} \right) \geq 1 - \delta,$$

where C is a constant, only depending on a, b, β .

2.3. Proof sketch for the pathwise regret bound (Theorem 3)

Here, we outline the main ideas underlying the proof of the pathwise regret bound, deferring the proofs of the technical lemmas to Appendix A.2. Throughout this section, we write

$V := \log(b/a)$ and $\delta_k := a(e^{V/k} - 1)$ for each $k \in \mathbb{N}$. We also write $\text{Hist}_{k,\downarrow}$ to denote the class of monotone histograms with k bins, that is

$$\text{Hist}_{k,\downarrow} := \left\{ h : x \mapsto \sum_{i=1}^k \theta_i \mathbb{1}_{\{t_{i-1} \leq x < t_i\}} \text{ for some } t_0, t_1, \dots, t_k \text{ and } \theta_1, \dots, \theta_k \right\} \cap \mathcal{D}_{a,b}.$$

The proof proceeds by comparing the EA algorithm to a monotone histogram approximation of the constrained Grenander estimator. We summarize the key steps below.

Step 1: a monotone histogram approximation of $\tilde{f}_{n,a,b}^{\text{MLE}}$ with controlled height variation: We begin by noting that, as in the unconstrained case, the constrained variant $\tilde{f}_{n,a,b}^{\text{MLE}}$ is also a monotone histogram. For the sake of completion, we state and prove this structural property in Lemma 10. However, there is no immediate control on the heights or breakpoints of $\tilde{f}_{n,a,b}^{\text{MLE}}$. Therefore, next, we establish such a control. To that end, first for each k , define the buffered histogram $\tilde{f}_{n,a+\delta_k,b}^{\text{MLE}}$. We start by proving that the buffered MLE is close to the original benchmark. For any $f \in \mathcal{D}_{a,b}$, define

$$T_{\delta_k}(f) := a + \delta_k + \frac{1 - a - \delta_k}{1 - a}(f - a).$$

Then $T_{\delta_k}(f) \in \mathcal{D}_{a+\delta_k,b}$ and $\sup_{x \in [0,1]} |T_{\delta_k}(f)(x) - f(x)| \leq C_1(a,b)\delta_k$. Further, it follows that

$$\sup_{x \in [0,1]} |\log T_{\delta_k}(f)(x) - \log f(x)| \leq C_2(a,b)\delta_k.$$

Applying this transformation to $f = \tilde{f}_{n,a,b}^{\text{MLE}}$ and using the optimality of $\tilde{f}_{n,a+\delta_k,b}^{\text{MLE}}$ gives

$$0 \leq \mathcal{L}(\tilde{f}_{n,a+\delta_k,b}^{\text{MLE}}, n) - \mathcal{L}(\tilde{f}_{n,a,b}^{\text{MLE}}, n) \leq C(a,b)n\delta_k.$$

Specifically, we show that $\tilde{f}_{n,a+\delta_k,b}^{\text{MLE}}$ can be approximated by a monotone histogram with at most k bins, while incurring only an additional cost of $O(1/k)$. We also show that despite using a compressed representation with at most k bins, adjacent bin heights are well separated on the logarithmic scale and the induced distortion in $\mathcal{L}(\cdot, n)$ is small. The following lemma formally states this.

Lemma 5 *Let $\tilde{f}_{n,a,b}^{\text{MLE}}$ and $\tilde{f}_{n,a+\delta_k,b}^{\text{MLE}}$ be the constrained Grenander estimators as defined in (3). For all sufficiently large integer $k \geq 1$, there exists a monotone histogram $h \in \bigcup_{r=1}^k \text{Hist}_{r,\downarrow}$ of the form*

$$h(x) = \sum_{i=1}^L \theta_i \mathbb{1}_{\{t_{i-1} \leq x < t_i\}}, \quad 0 = t_0 < t_1 < \dots < t_L = 1, \quad L \leq k,$$

such that $t_{L-1} \leq \max_{1 \leq i \leq n} X_i$ and, for every $j \in \{1, \dots, L-1\}$,

$$\log \theta_j - \log \theta_{j+1} \geq \frac{V}{k}. \quad (8)$$

Moreover, h satisfies

$$|\mathcal{L}(\tilde{f}_{n,a+\delta_k,b}^{\text{MLE}}, n) - \mathcal{L}(h, n)| \leq 2n \frac{V}{k}. \quad (9)$$

Since we have proved $0 \leq \mathcal{L}(\tilde{f}_{n,a+\delta_k,b}^{\text{MLE}}, n) - \mathcal{L}(\tilde{f}_{n,a,b}^{\text{MLE}}, n) \leq C(a,b)n\delta_k$, we consequently obtain

$$\mathcal{L}(h, n) - \mathcal{L}(\tilde{f}_{n,a,b}^{\text{MLE}}, n) \leq \frac{2nV}{k} + C(a,b)n\delta_k.$$

We denote by $\tilde{f}_{n,(k),a,b}^{\text{MLE}}$ the compressed histogram returned by Lemma 5, with breakpoints $0 = t_0 < t_1 < \dots < t_L = 1$ and heights $\theta_1 \geq \dots \geq \theta_L$. It is obtained by compressing $\tilde{f}_{n,a+\delta_k,b}^{\text{MLE}}$, but belongs to the original class $\mathcal{D}_{a,b}$. This approximation serves as the offline benchmark against which the EA algorithm is compared in the remainder of the proof.

Step 2: Discretizing the compressed estimator into the expert class. We now show that the compressed histogram $\tilde{f}_{n,(k),a,b}^{\text{MLE}}$ constructed in Step 1 can be closely approximated by an element of the expert class $\mathcal{E}_{k,n,\beta}$, with only a small additional distortion in $\mathcal{L}(\cdot, n)$.

The key ingredient is the following lemma, which shows that any monotone histogram with well-separated log-heights can be rounded to the expert grid without significantly affecting $\mathcal{L}(\cdot, n)$. While this may seem to follow from a straightforward rounding of endpoints and heights, the proof requires careful arguments to preserve feasibility, monotonicity, and continued membership in $\mathcal{D}_{a,b}$.

Lemma 6 *Fix integers $n \geq 1$ and $k \geq 2$. Let $f \in \bigcup_{r=1}^k \text{Hist}_{r,\downarrow}$ be of the form*

$$f(x) = \sum_{j=1}^r \theta_j \mathbb{1}\{t_{j-1} \leq x < t_j\}, \quad 0 = t_0 < t_1 < \dots < t_r = 1, \quad r \leq k,$$

and suppose that $t_{r-1} \leq 1 - n^{-\gamma}$ and $\log \theta_{r-1} - \log \theta_r \geq V/k$. Then there exists a histogram $g \in \mathcal{E}_{k,n,\beta}$ whenever

$$\frac{V}{k} < 1, \quad V\Delta_b < 1, \quad a\frac{V}{k} \geq b\Delta_b V + b(\Delta_b V + 2k\Delta_b)n^\gamma. \quad (10)$$

Moreover, for any $(x_1, \dots, x_n) \in \mathcal{S}_n(\beta, \gamma)$, we have that $\mathcal{L}(g, n) - \mathcal{L}(f, n) \leq n\Delta_b V + (k-1)V$.

Here, we apply Lemma 6 with $f = \tilde{f}_{n,(k),a,b}^{\text{MLE}}$. Note that by Lemma 5 and the assumption that $(x_1, \dots, x_n) \in \mathcal{S}_n(\beta, \gamma)$, the final breakpoint satisfies $t_{L-1} \leq \max_{1 \leq i \leq n} x_i \leq 1 - n^{-\gamma}$, and that the last two heights obey $\log \theta_{L-1} - \log \theta_L \geq \frac{V}{k}$. Therefore, whenever (10) holds, there exists a histogram $g_{n,k,\beta}^* \in \mathcal{E}_{k,n,\beta}$ and

$$\mathcal{L}(g_{n,k,\beta}^*, n) - \mathcal{L}(\tilde{f}_{n,(k),a,b}^{\text{MLE}}, n) \leq n\Delta_b V + (k-1)V.$$

Step 3: bounding the size of the expert class $\mathcal{E}_{k,n,\beta}$. We now derive an upper bound on the cardinality of the expert class $\mathcal{E}_{k,n,\beta}$ in terms of n , k , and β . By definition, the grid of admissible breakpoints satisfies

$$|\mathcal{B}_n| \leq n^{\beta+1} + 2.$$

The discretization of the log-heights is defined as $\Lambda_{n,a,b} := \log(a) + V\mathcal{B}_n$, so that $|\Lambda_{n,a,b}| = |\mathcal{B}_n|$.

We upper bound the size of $\mathcal{E}_{k,n,\beta}$ by ignoring the monotonicity constraint on the heights and the feasibility conditions imposed on the last height, since both can only reduce the number of admissible histograms. A histogram in $\mathcal{E}_{k,n,\beta}$ with at most k bins is specified by at most $k-1$ internal breakpoints chosen from $\mathcal{B}_n$ and at most $k-1$ independent log-height values chosen from $\Lambda_{n,a,b}$. Hence,

$$|\mathcal{E}_{k,n,\beta}| \leq |\mathcal{B}_n|^{k-1} |\Lambda_{n,a,b}|^{k-1} \lesssim (n^{\beta+1})^{2k-2},$$

where the implicit constant is universal. Consequently, $\log |\mathcal{E}_{k,n,\beta}| \lesssim k(\beta+1) \log n$.

Step 4: Aggregation over the finite expert class. We now combine the previous steps and complete the proof of regret bound for EA algorithm $\hat{f}_{a,b}^{\text{EA}}$ defined over the finite class $\mathcal{E}_{k,n,\beta}$. The key property we use is that exponential aggregation under log-loss performs nearly as well as the best expert in hindsight, up to a logarithmic penalty in the size of the class.

We state and prove this in Lemma 13, by which we have that $\hat{f}_{a,b}^{\text{EA}}$ satisfies

$$\mathcal{L}(\hat{f}_{a,b}^{\text{EA}}, n) \leq \min_{g \in \mathcal{E}_{k,n,\beta}} \mathcal{L}(g, n) + \log |\mathcal{E}_{k,n,\beta}|. \quad (11)$$

From Step 3, we have $\log |\mathcal{E}_{k,n,\beta}| \lesssim k(\beta+1) \log n$. Moreover, combining Steps 1 and 2, whenever the condition (10) holds, there exists an expert $g_{n,k,\beta}^* \in \mathcal{E}_{k,n,\beta}$ such that

$$\mathcal{L}(g_{n,k,\beta}^*, n) - \mathcal{L}(\tilde{f}_{n,a,b}^{\text{MLE}}, n) \leq C(a, b)n\delta_k + 2n\frac{V}{k} + nV\Delta_b + (k-1)V.$$

Since $\delta_k = a(e^{V/k} - 1) = O(1/k)$, the additional term $n\delta_k$ is of order n/k . Now, we make the choice $k \asymp \sqrt{\frac{n}{\log n}}$, and recall that $\Delta_b = n^{-(\beta+1)}$ with $\beta > 0$. With this choice, the right-hand side above is of order $\sqrt{n \log n}$, and the size of $\mathcal{E}_{k,n,\beta}$ also satisfies $\log |\mathcal{E}_{k,n,\beta}| \lesssim \sqrt{n \log n}$. Furthermore, the condition (10) is satisfied provided that $\gamma < \beta - \frac{1}{2}$. Substituting these bounds into (11), we conclude that

$$\text{Regret}(\hat{f}_{a,b}^{\text{EA}}; n, a, b) = \mathcal{L}(\hat{f}_{a,b}^{\text{EA}}, n) - \mathcal{L}(\tilde{f}_{n,a,b}^{\text{MLE}}, n) \lesssim \sqrt{n \log n},$$

uniformly over all sequences $(x_1, \dots, x_n) \in \mathcal{S}_n(\beta, \gamma)$. This completes the proof of Theorem 3.

3. Applying monotone density estimation to sequential testing

This section shows how online monotone density estimation can be used to construct sequential tests for a fixed null hypothesis H_0 . We begin with a brief review of sequential testing and the role of e-values as a core tool in developing these tests. We then introduce p-to-e calibration as a way to embed classical p-value based tests into the e-value framework, and show that learning ‘optimal’ calibrators naturally reduces to online monotone density estimation. Finally, we establish theoretical guarantees for the resulting calibrators.

3.1. Sequential testing: background

Sequential testing studies hypothesis testing with data observed sequentially, allowing decisions to be made at data-dependent times. Such problems arise naturally in modern machine learning and statistics, and have a long history in statistics (Barnard, 1946; Wald, 1992; Gurevich et al., 2019).

E-values have emerged as a convenient and flexible tool for constructing sequential tests in this setting (Wasserman et al., 2020; Vovk and Wang, 2021; Grünwald et al., 2024; Ramdas et al., 2023); see Ramdas and Wang (2025) for a comprehensive treatment. Formally, an e-value is a nonnegative random variable E satisfying $\mathbb{E}[E] \leq 1$ under the null hypothesis. This simple expectation constraint underlies their use in tests that remain valid under data-dependent stopping rules.

In a sequential setting, suppose $(E_t)_{t \geq 1}$ is a sequence of e-values adapted to a filtration $(\mathcal{F}_t)$, with $\mathbb{E}_{H_0}[E_t \mid \mathcal{F}_{t-1}] \leq 1$ for all $t \geq 1$. An associated e-process $M = (M_t)_{t \geq 1}$ can be defined as

$$M_t = \prod_{s=1}^t E_s, \quad M_0 = 1. \quad (12)$$

The process (M_t) is a nonnegative supermartingale under the null hypothesis. As a consequence, for any data-dependent stopping time τ satisfying $\{\tau \leq t\} \in \mathcal{F}_t$, we have $\mathbb{E}_{H_0}[M_\tau] \leq 1$, and we also have $\mathbb{P}_{H_0}(\sup_{t \geq 1} M_t \geq \frac{1}{\alpha}) \leq \alpha$ for all $\alpha \in (0, 1]$. Therefore, thresholding (M_t) at $1/\alpha$ yields a valid sequential test of H_0 , regardless of the stopping rule. These properties form the basis for widespread use of e-values as building blocks for sequential inference.

On the other hand, most classical hypothesis testing procedures are formulated in terms of p-values and are designed for fixed-sample settings. While supported by a rich theory characterizing their validity and power in offline regimes, such tests generally fail to control Type I error under data-dependent stopping rules. To leverage this extensive body of work in sequential settings, it is therefore natural to seek principled ways to embed p-value based methods into the e-value framework. *p-to-e calibration* provides a simple and general mechanism for doing so.

3.2. Embedding classical tests via p-to-e calibration

Formally, a p-to-e calibrator is a function that maps p-values to e-values. That is, for any valid p-value P , the transformed quantity $h(P)$ must be a valid e-value. Since this requirement must hold uniformly over all super-uniform p-values, it imposes a structural constraint on the calibrator: it must be dominated by a nonnegative and non-increasing function on $[0, 1]$.

One defines a *p-to-e calibrator* as a non-increasing function $h : [0, 1] \rightarrow [0, \infty)$ satisfying $\int_0^1 h(p) dp \leq 1$, equivalently, a monotone (sub-)probability Lebesgue-density function on $[0, 1]$. By definition, if P is a valid p-value, then $h(P)$ is a valid e-value. Vovk and Wang (2021) further show that any admissible calibrator must attain equality. Thus, we observe that

All admissible calibrators are decreasing densities supported on $[0, 1]$.

We restrict our attention to the admissible calibrators, that is the class $\mathcal{D}$. The connection between calibration and monotone densities plays a central role in our development of the calibrators; this link has appeared previously in [Arnold et al. \(2023\)](#). From this point onward, we refer to a p-to-e calibrator simply as a calibrator.

We now describe how calibrators integrate with the e-value framework when only p-values are observed. Suppose we observe a sequence $(P_t)_{t \geq 1}$ of p-variables produced by classical hypothesis tests, satisfying under the null hypothesis H_0 that

$$\mathbb{P}_{H_0}(P_t \leq \alpha \mid \mathcal{F}_{t-1}) \leq \alpha, \quad \text{for all } t \geq 1, \alpha \in (0, 1).$$

Our goal is to convert these p-values into e-values in an online and adaptive manner. At each time t , we select an $\mathcal{F}_{t-1}$ -predictable calibrator $h_t \in \mathcal{D}$ and transform P_t into an e-value $h_t(P_t)$. The resulting e-process is given by

$$M_t = \prod_{s=1}^t h_s(P_s), \quad M_0 = 1. \quad (13)$$

By construction, this enables a valid sequential test for any such online calibrator.

In the conformal prediction literature, non-increasing p-to-e calibrators are known as betting functions, and the e-process (13) is a conformal test martingale when the underlying sequential conformal p-values are uniform under the null. Thus, our adaptive calibrators may equivalently be viewed as predictable betting functions for constructing conformal test martingales. In the next section, we show how online monotone density estimation provides a systematic method for learning such betting functions, yielding conformal test martingales with concrete theoretical guarantees, including asymptotic log-optimality.

3.3. Empirically adaptive p-to-e calibration

While any calibrator h yields a valid sequential test via (13), the power of the resulting procedure depends critically on how the calibrators are chosen. To formalize an optimal benchmark, [Ramdas and Wang \(2025, Section 3.6\)](#) introduce the following oracle procedure.

Definition 7 (Log-optimal decreasing calibrator) *For a given alternative distribution $\mathbb{Q}$, the $\mathbb{Q}$ -log-optimal calibrator $h^{\text{opt}} = (h_t^{\text{opt}})_{t \geq 1}$ is defined by*

$$h_t^{\text{opt}} \in \operatorname{argmax}_{h \in \mathcal{D}} \mathbb{E}_{\mathbb{Q}}[\log h(P_t) \mid \mathcal{F}_{t-1}].$$

Since the alternative distribution $\mathbb{Q}$ is unknown, the oracle calibrator h_t^{opt} is not directly computable. This motivates the construction of empirically adaptive calibrators that approximate the oracle and achieve high power in practice.

Throughout the remainder of this section, we work under the alternative hypothesis, which posits that $P_1, P_2, \dots \stackrel{iid}{\sim} \mathbb{Q}$ with a monotone density $q \in \mathcal{D}$. The i.i.d. model considered here serves as a natural starting point for the study of adaptive p-to-e calibration. While practical sequential testing problems often involve dependence and non-stationarity, empirically adaptive p-to-e calibration remains relatively underexplored even in the iid setting. Moreover, under most alternatives arising in standard testing problems, p-values tend

to concentrate near zero and exhibit a decreasing density, aligning naturally with our modeling assumption. For instance, in one-sided Gaussian mean testing, $H_0 : \mu = 0$ versus $H_1 : \mu > 0$ with $Z \sim N(\mu, 1)$, the corresponding p -value $\Phi(-Z)$ has a monotone decreasing density under the alternative.

In the above iid model, the log-optimal calibrator is given by $h_t^{\text{opt}} = q$ for all $t \geq 1$. Indeed, for any $h \in \mathcal{D}$,

$$\mathbb{E}_{\mathbb{Q}}[\log h(P_t) \mid \mathcal{F}_{t-1}] \leq \mathbb{E}_{\mathbb{Q}}[\log q(P_t) \mid \mathcal{F}_{t-1}] - \text{KL}(q \parallel h) \leq \mathbb{E}_{\mathbb{Q}}[\log q(P_t) \mid \mathcal{F}_{t-1}].$$

Consequently, we obtain the following interesting fact, which appears implicitly in prior work but has not been systematically developed:

The log-optimal calibrator is given by q . Therefore, approximating the log-optimal calibrator from the observed p -values reduces to the problem of online monotone density estimation.

We therefore use the two online monotone density estimators introduced in Section 2 as empirically adaptive calibrators:

- **OG calibrator.** The estimator $\hat{f}_{a,b}^{\text{OG}}$ yields a natural adaptive calibrator. Its online likelihood maximization can be interpreted as empirical log-optimal calibration, where the conditional distribution of the p -value P_t under $\mathbb{Q}$ is replaced by the empirical distribution of past observations, $(t-1)^{-1} \sum_{s=1}^{t-1} \delta_{P_s}$.
- **EA calibrator.** Alternatively, $\hat{f}_{a,b}^{\text{EA}}$ defines an adaptive calibrator by exponentially weighting a finite collection of expert calibrators, with weights updated according to accumulated log-wealth from past p -values.

Both $\hat{f}_{a,b}^{\text{OG}}$ and $\hat{f}_{a,b}^{\text{EA}}$ yield valid e -processes via (13), and therefore induce valid sequential tests. In practice, a natural choice is to take $a > 0$ and $b = \infty$ to ensure that calibrators never output zero while allowing unbounded values to enable strong evidence accumulation.

3.4. Optimality of empirically adaptive calibrators

We now study optimality properties of these empirically adaptive calibrators, establishing their effectiveness as practical procedures for sequential testing. In particular, we show that both the OG and EA algorithms are asymptotically log-optimal calibrators.

Theorem 8 (Asymptotic log-optimality) *Suppose $P_1, P_2, \dots \stackrel{\text{iid}}{\sim} \mathbb{Q}$, where $q \in \mathcal{D}_{a,b}$ for some $0 < a < b < \infty$. Then for any calibrator $\hat{h} \in \{\hat{f}_{a,b}^{\text{OG}}, \hat{f}_{a,b}^{\text{EA}}\}$,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \left(\log M_n(\hat{h}) - \log M_n(h^{\text{opt}}) \right) = 0 \quad \mathbb{Q}\text{-almost surely,}$$

where h^{opt} denotes the log-optimal calibrator from Definition 7.

Here, by $\hat{f}_{a,b}^{\text{EA}}$ we refer to the EA calibrator based on the set of experts $\mathcal{E}_{k,n,\beta}$ with $k = \lfloor (n/\log n)^{1/2} \rfloor$. Theorem 8 shows that the proposed calibrators asymptotically achieve the same per-sample log-wealth as the optimal calibrator tailored to the true distribution $\mathbb{Q}$. As a consequence, for both procedures, the accumulated evidence diverges under any non-uniform monotone alternative.

Corollary 9 (Almost sure divergence of log-wealth) Suppose $P_1, P_2, \dots \stackrel{iid}{\sim} \mathbb{Q}$, where $q \in \mathcal{D}_{a,b}$ for some $0 < a < b < \infty$ and $\int_0^1 q(u) \log q(u) du > 0$. Then for any $\hat{h} \in \{\hat{f}_{a,b}^{\text{OG}}, \hat{f}_{a,b}^{\text{EA}}\}$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log M_n(\hat{h}) = \int_0^1 q(u) \log q(u) du \quad \mathbb{Q}\text{-almost surely.}$$

In particular, $\log M_n(\hat{h}) \rightarrow \infty$ $\mathbb{Q}$ -almost surely as $n \rightarrow \infty$.

We observe that the condition $\text{KL}(q \| f_U) = \int_0^1 q(u) \log q(u) du > 0$, where f_U denotes the density of $\text{Unif}[0, 1]$, simply requires $\mathbb{Q} \not\stackrel{d}{=} \text{Unif}[0, 1]$. Under such alternatives, we do expect the log-wealth to diverge and the associated sequential test to have a finite detection time almost surely.

4. Numerical experiments

Now, we study the empirical performance of the proposed online estimators through a series of numerical experiments.

4.1. Empirical evaluation of OG and EA in online monotone density estimation

We compare the cumulative log-likelihood trajectories of the OG and EA algorithms against two benchmarks: the true density q and the offline Grenander estimator $\tilde{f}_n^{\text{MLE}}$ computed in hindsight. We consider sample paths of length $n = 1000$ and report averages over $B = 50$ independent replications. For the EA procedure, we use a structured collection of monotone histogram experts with 2–4 breakpoints chosen from a coarse grid on $(0.1, 0.2, \dots, 0.9)$ and geometrically decaying heights, normalized to integrate to one. Although substantially smaller than the theoretical construction in Section 2.2, this expert class remains sufficiently expressive in practice and performs well empirically.

Well-specified stochastic setting. We start by considering the well-specified regime, where $X_1, X_2, \dots \stackrel{iid}{\sim} \mathbb{Q}$ with a monotone density $q \in \mathcal{D}$. We examine three representative models: (1) a linear density $q(u) = 5/4 - u/2$, (2) a quadratic density $q(u) = 3(1 - u)^2$, and (3) a piecewise-constant density on four equal-width bins with heights $5/4, 13/12, 11/12$, and $3/4$.

Although our theoretical guarantees assume that the decreasing density is uniformly bounded above and bounded away from zero, the proposed OG and EA algorithms can also be applied beyond this setting. In this experiment, we include the quadratic model, whose density vanishes at the boundary ($u = 1$), to empirically demonstrate this.

Figure 1 reports the resulting cumulative log-likelihood trajectories. As expected, the offline Grenander estimator is optimistic and attains the largest cumulative likelihood. Both online procedures closely track the true density, while EA typically exhibits smaller regret relative to the offline Grenander benchmark, indicating improved finite-sample adaptivity. This difference is most pronounced in the linear model, where the monotone structure is relatively mild; in the remaining settings, the two methods behave similarly.

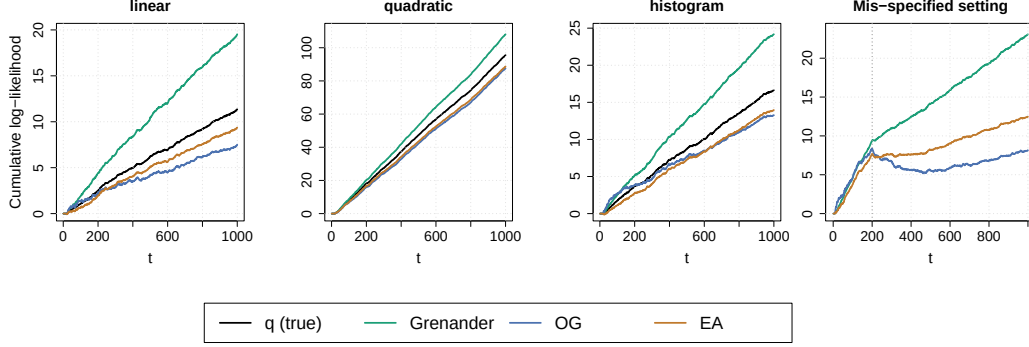


Figure 1: Cumulative log-likelihood trajectories in well-specified and mis-specified stochastic settings.

Mis-specified setting. We next consider a mis-specified scenario in which the data-generating density changes over time. Specifically, observations are initially drawn iid from the linear monotone density $q(u) = \delta_1 - (\delta_1 - \delta_0)u$ with $(\delta_0, \delta_1) = (0.5, 1.5)$ up to a change point $\tau = 200$, after which the parameters switch to $(\delta_0, \delta_1) = (0.75, 1.25)$.

The rightmost panel of Figure 1 illustrates the resulting cumulative log-likelihood trajectories. In this setting, EA empirically adjusts more rapidly to the evolving density and incurs substantially smaller regret. This suggests that aggregation over the expert class makes EA less strongly influenced by the pre-change observations than OG. In contrast, OG is more strongly influenced by pre-change observations and therefore adapts more slowly following the shift. This experiment highlights the advantage of expert aggregation when the monotone or iid assumptions are only approximately satisfied.

4.2. Sequential changepoint detection

We now study the performance of the OG and EA calibrators in a sequential changepoint detection problem. We simulate a sequence $X_1, X_2, \dots, X_n$ with $n = 2000$, where

$$X_t \sim N(0, 1) \quad \text{for } t \leq \tau, \quad X_t \sim N(\mu, 1) \quad \text{for } t > \tau,$$

with changepoint $\tau = 500$. At each time step $t \in [n]$, we construct conformal p -values as

$$p_t = \frac{1}{t} \left(\sum_{i < t} \mathbb{1}\{X_i > X_t\} + U_t \sum_{i=1}^t \mathbb{1}\{X_i = X_t\} \right), \quad (U_1, \dots, U_n) \stackrel{iid}{\sim} \text{Unif}[0, 1].$$

This provides one of the simplest conformal p -value constructions in this setting. Observe that for $t \leq \tau$, the sequence $(X_1, \dots, X_t)$ is exchangeable, and hence $p_t \sim \text{Unif}[0, 1]$. In fact, p_t is conditionally uniform with respect to the filtration generated by the past p -values; that is,

$$\mathbb{P}(p_t \leq \alpha \mid \mathcal{F}(p_1, \dots, p_{t-1})) = \alpha, \quad \text{for } \alpha \in [0, 1].$$

After the changepoint, however, the observations become systematically larger; this forces the conformal p -values to concentrate near zero and, marginally, exhibit a near-decreasing density.

The resulting sequence of conformal p -values is incorporated into (13) to construct sequential test martingales. A classical approach in the conformal martingale literature is to use a fixed calibrator, that is, $h_s = h$ for all $s \in [n]$; see Vovk et al. (2003); Volkhonskiy et al. (2017). Following Volkhonskiy et al. (2017, Section 3.6), we consider the following two fixed baseline calibrators:

- **Mixture-power calibrator:**

$$h_{\text{mix}}(p) = \int_0^1 \varepsilon p^{\varepsilon-1} d\varepsilon,$$

which corresponds to a continuous mixture of power betting functions.

- **Constant-step calibrator:**

$$h_{\text{step}}(p) = \begin{cases} 1.1 & \text{if } p \leq 0.5 \\ 0.9 & \text{otherwise.} \end{cases}$$

We compare these two fixed-calibrator baselines with the adaptive OG and EA calibrators. For the EA procedure, we use the same expert class as in Section 4.1.

We consider three choices of the post-change mean, namely $\mu \in \{0.5, 1, 2\}$, corresponding respectively to weak, moderate, and strong mean shifts, thereby ranging from harder to easier changepoint detection problems. For each setting, we generate 100 independent repetitions and report the average cumulative log-wealth trajectories of the competing test martingales in Figure 2.

Discussion on the results. Across all settings, both the OG and EA calibrators perform robustly, and the corresponding test martingales eventually cross the rejection threshold, correctly detecting the change in the data-generating process. As expected, weaker shifts require substantially longer detection times.

In contrast, the performance of the fixed calibrators heavily depends on the signal strength. In the strong-shift regime ($\mu = 2$), the post-change p -values become extremely small, which is effectively exploited by the mixture-power calibrator (denoted as CM_mix in Figure 2), making it competitive with the OG and EA procedures and substantially stronger than the constant-step calibrator. On the other hand, in the weak-shift regime ($\mu = 0.5$), the post-change p -values deviate only mildly from uniformity. In this case, the constant-step calibrator (denoted as CM_const in Figure 2) performs more favorably than the mixture-power alternative.

We also observe that, since the sequence length is relatively large ($n = 2000$) compared to the changepoint location ($\tau = 500$), the conformal p -values gradually become closer to uniform at later time points. Consequently, the corresponding test martingales may lose wealth after an initial growth phase. This effect is particularly pronounced for the mixture-power calibrator.

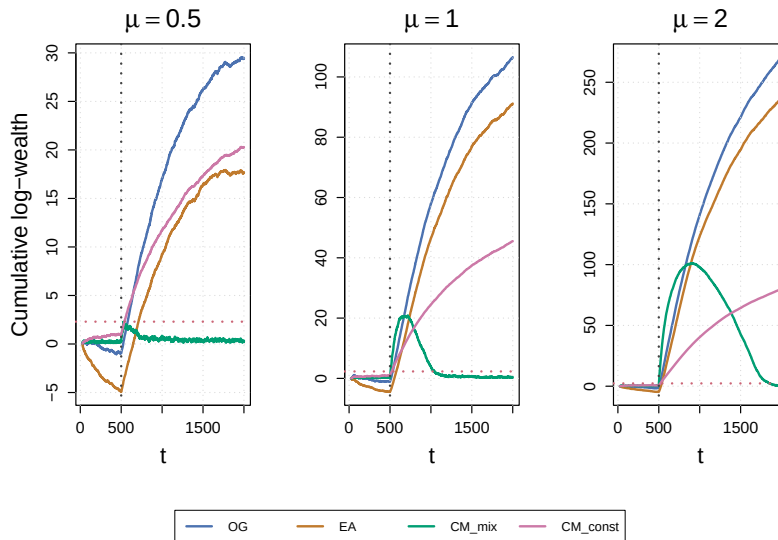


Figure 2: Average cumulative log-wealth trajectories for sequential changepoint detection under weak ($\mu = 0.5$), moderate ($\mu = 1$), and strong ($\mu = 2$) mean shifts, shown from left to right. The vertical dashed line marks the changepoint location $\tau = 500$, while the horizontal red line denotes the rejection threshold $\log(1/\alpha)$ with $\alpha = 0.1$.

5. Discussion

In this work, we introduced two online algorithms for monotone density estimation. In the well-specified stochastic setting, where the true density q is monotone, we derived bounds on the excess KL-risk of both procedures. We also obtained a pathwise regret bound for the EA algorithm, comparing its performance to that of the offline Grenander estimator computed in hindsight.

We further showed that the problem of p-to-e calibration in sequential testing can be naturally formulated as an instance of online monotone density estimation. This connection allowed us to adapt the proposed online algorithms as data-driven calibrators and to establish their optimality.

A promising direction for future work is to analyze the behavior of these empirically adaptive calibrators under non-iid alternatives, which would further enhance their applicability in practical sequential testing settings.

Acknowledgments

Rohan Hore and Aaditya Ramdas acknowledge the funding from the Sloan Fellowship. Ruodu Wang is supported by the Natural Sciences and Engineering Research Council of Canada (CRC-2022-00141, RGPIN-2024-03728). The authors also thank Aditya Guntuboyina and Johanna Ziegel for pointers to relevant literature.

References

- Sebastian Arnold, Alexander Henzi, and Johanna F Ziegel. Sequentially valid tests for forecast calibration. *The Annals of Applied Statistics*, 17(3):1909–1935, 2023.
- Fadoua Balabdaoui and Jon A Wellner. Estimation of a k -monotone density: characterizations, consistency and minimax lower bounds. *Statistica Neerlandica*, 64(1):45–70, 2010.
- George A Barnard. Sequential tests in industrial statistics. *Supplement to the Journal of the Royal Statistical Society*, 8(1):1–21, 1946.
- Lucien Birge. The Grenander estimator: A nonasymptotic approach. *The Annals of Statistics*, pages 1532–1549, 1989.
- Chris Carolan and Richard Dykstra. Asymptotic behavior of the Grenander estimator at density flat regions. *Canadian Journal of Statistics*, 27(3):557–566, 1999.
- Nicolo Cesa-Bianchi and Gábor Lugosi. *Prediction, learning, and games*. Cambridge University Press, 2006.
- Cécile Durot, Vladimir N Kulikov, and Hendrik P Lopuhaä. The limit distribution of the l_∞ -error of Grenander-type estimators. *The Annals of Statistics*, 2012.
- Fuchang Gao and Jon A Wellner. Entropy estimate for high-dimensional monotonic functions. *Journal of Multivariate Analysis*, 98(9):1751–1764, 2007.
- FuChang Gao and Jon A Wellner. On the rate of convergence of the maximum likelihood estimator of a k -monotone density. *Science in China Series A: Mathematics*, 52(7):1525–1538, 2009.
- Ulf Grenander. On the theory of mortality measurement: Part ii. *Scandinavian Actuarial Journal*, 1956(2):125–153, 1956.
- Piet Groeneboom and HP Lopuhaa. Isotonic estimators of monotone densities and distribution functions: basic facts. *Statistica Neerlandica*, 47(3):175–183, 1993.
- Piet Groeneboom, Gerard Hooghiemstra, and Hendrik P Lopuhaä. Asymptotic normality of the l_1 error of the Grenander estimator. *The Annals of Statistics*, 27(4):1316–1347, 1999.
- Peter Grünwald, Rianne De Heide, and Wouter Koolen. Safe testing. *Journal of the Royal Statistical Society Series B (with Discussion)*, 86(5):1091–1128, 2024.
- Andrey Gurevich, Kobi Cohen, and Qing Zhao. Sequential anomaly detection under a nonlinear system cost. *IEEE Transactions on Signal Processing*, 67(14):3689–3703, 2019.
- Shuji Hao, Peiying Hu, Peilin Zhao, Steven CH Hoi, and Chunyan Miao. Online active learning with expert advice. *ACM Transactions on Knowledge Discovery from Data (TKDD)*, 12(5):1–22, 2018.

- Vladimir N. Kulikov and Hendrik P. Lopuhaä. Asymptotic normality of the L_k -error of the Grenander estimator. *The Annals of Statistics*, 33(5):2228 – 2255, 2005. doi: 10.1214/009053605000000462. URL <https://doi.org/10.1214/009053605000000462>.
- Aaditya Ramdas and Ruodu Wang. Hypothesis testing with e-values. *Foundations and Trends® in Statistics*, 1(1-2):1–390, 2025.
- Aaditya Ramdas, Peter Grünwald, Vladimir Vovk, and Glenn Shafer. Game-theoretic statistics and safe anytime-valid inference. *Statistical Science*, 38(4):576–601, 2023.
- BLS Prakasa Rao. Estimation of a unimodal density. *Sankhyā: The Indian Journal of Statistics, Series A*, pages 23–36, 1969.
- Sara Van de Geer. Hellinger-consistency of certain nonparametric maximum likelihood estimators. *The Annals of Statistics*, 21(1):14–44, 1993.
- Denis Volkhonskiy, Evgeny Burnaev, Ilia Nouretdinov, Alexander Gammernan, and Vladimir Vovk. Inductive conformal martingales for change-point detection. In *Conformal and Probabilistic Prediction and Applications*, pages 132–153. PMLR, 2017.
- Vladimir Vovk and Ruodu Wang. E-values: Calibration, combination, and applications. *The Annals of Statistics*, 49:1736–1754, 2021.
- Vladimir Vovk, Ilia Nouretdinov, and Alexander Gammernan. Testing exchangeability online. In *Proceedings of the 20th international conference on machine learning (ICML-03)*, pages 768–775, 2003.
- Abraham Wald. Sequential tests of statistical hypotheses. In *Breakthroughs in statistics: Foundations and basic theory*, pages 256–298. Springer, 1992.
- Larry Wasserman, Aaditya Ramdas, and Sivaraman Balakrishnan. Universal inference. *Proceedings of the National Academy of Sciences*, 117(29):16880–16890, 2020.
- Wing Hung Wong and Xiaotong Shen. Probability inequalities for likelihood ratios and convergence rates of sieve MLEs. *The Annals of Statistics*, pages 339–362, 1995.

Appendix A. Proofs

A.1. Proof of results from Section 2

A.1.1. PROOF OF LEMMA 1

Fix $0 < a < b < \infty$ throughout the proof. We write $N_{[]} (u, \mathcal{D}_{a,b}, d_H)$ for the bracketing entropy of $\mathcal{D}_{a,b}$ with respect to the Hellinger distance. By [Wong and Shen \(1995, Theorem 1\)](#), there exist universal constants $c_1, c_2 > 0$ such that for any $\varepsilon > 0$ satisfying

$$\int_0^\varepsilon \sqrt{N_{[]} (u, \mathcal{D}_{a,b}, d_H)} du \leq c_1 \sqrt{n} \varepsilon^2, \quad (14)$$

we have $\mathbb{P}_{\mathbb{Q}}(\mathcal{A}) \leq 2 \exp(-c_2 n \varepsilon^2)$, where

$$\mathcal{A} := \left\{ \sup_{f \in \mathcal{D}_{a,b}: d_H(f,q) \geq \varepsilon} \sum_{s=1}^n \log \frac{f(X_s)}{q(X_s)} \geq -c_2 n \varepsilon^2 \right\}. \quad (15)$$

By definition of the MLE,

$$\sum_{s=1}^n \log \tilde{f}_{n,a,b}^{\text{MLE}}(X_s) \geq \sum_{s=1}^n \log q(X_s),$$

and hence

$$\sum_{s=1}^n \log \frac{\tilde{f}_{n,a,b}^{\text{MLE}}(X_s)}{q(X_s)} \geq 0.$$

On the event $\mathcal{A}^c$, any density f satisfying $d_H(f, q) \geq \varepsilon$ obeys

$$\sum_{s=1}^n \log \frac{f(X_s)}{q(X_s)} < -c_2 n \varepsilon^2 < 0,$$

and therefore cannot coincide with the MLE $\tilde{f}_{n,a,b}^{\text{MLE}}$. Consequently,

$$\mathcal{A}^c \subseteq \{d_H(\tilde{f}_{n,a,b}^{\text{MLE}}, q) < \varepsilon\}.$$

Thus, for any $\varepsilon > 0$ satisfying (14),

$$\mathbb{P}_{\mathbb{Q}}(d_H(\tilde{f}_{n,a,b}^{\text{MLE}}, q) \geq \varepsilon) \leq 2 \exp(-c_2 n \varepsilon^2).$$

By [Gao and Wellner \(2007\)](#), the bracketing entropy of $\mathcal{D}_{a,b}$ in Hellinger distance satisfies

$$N_{[]} (u, \mathcal{D}_{a,b}, d_H) \leq C u^{-1},$$

for a constant $C > 0$ depending only on a and b . Therefore,

$$\int_0^\varepsilon \sqrt{N_{[]} (u, \mathcal{D}_{a,b}, d_H)} du \leq C \int_0^\varepsilon u^{-1/2} du = 2C \varepsilon^{1/2}.$$

The entropy condition (14) thus holds for all $\varepsilon \geq \varepsilon_n$, where $\varepsilon_n \asymp n^{-1/3}$. It follows that

$$\mathbb{E}_{\mathbb{Q}}[d_H^2(\tilde{f}_{n,a,b}^{\text{MLE}}, q)] \leq \varepsilon_n^2 + \int_{\varepsilon_n^2}^1 2 \exp(-c_2 n s) ds \leq \varepsilon_n^2 + \frac{2}{c_2 n} \leq C'(a, b) n^{-2/3},$$

for an appropriate constant $C'(a, b) > 0$. Finally, by Lemma 12, since $q, \tilde{f}_{n,a,b}^{\text{MLE}} \in \mathcal{D}_{a,b}$,

$$\text{KL}(q \| \tilde{f}_{n,a,b}^{\text{MLE}}) \leq C''(a, b) d_H^2(q, \tilde{f}_{n,a,b}^{\text{MLE}})$$

for another constant $C''(a, b) > 0$. Taking expectations yields

$$\mathbb{E}_{\mathbb{Q}}[\text{KL}(q \| \tilde{f}_{n,a,b}^{\text{MLE}})] \leq \kappa(a, b) n^{-2/3},$$

for some constant $\kappa(a, b) \in (0, \infty)$, as required. ■

A.1.2. PROOF OF THEOREM 2: BOUNDS ON EXCESS KL-RISK

Fix $n \geq 2$ and $0 < a < b < \infty$ throughout.

Bounding the risk of the OG estimator. We first establish the risk bound for the online Grenander (OG) estimator. For any $t \geq 2$, note that

$$\hat{f}_{t,a,b}^{\text{OG}} = \tilde{f}_{t-1,a,b}^{\text{MLE}}.$$

Recall that

$$\text{Risk}_{\mathbb{Q}}(\hat{f}; n) = \sum_{s=1}^n \mathbb{E}_{\mathbb{Q}}[\text{KL}(q \| \hat{f}_s)].$$

Thus, by Lemma 1, for any $q \in \mathcal{D}_{a,b}$,

$$\begin{aligned} \text{Risk}_{\mathbb{Q}}(\hat{f}_{a,b}^{\text{OG}}; n) &= \sum_{s=1}^n \mathbb{E}_{\mathbb{Q}}[\text{KL}(q \| \hat{f}_{s,a,b}^{\text{OG}})] \\ &= \sum_{s=2}^n \mathbb{E}_{\mathbb{Q}}[\text{KL}(q \| \tilde{f}_{s-1,a,b}^{\text{MLE}})] + \mathbb{E}_{\mathbb{Q}}[\text{KL}(q \| 1)] \\ &\leq \kappa_{a,b} \sum_{s=2}^n (s-1)^{-2/3} + \text{KL}(q \| 1) \leq \Gamma_{\text{OG}}(a, b) n^{1/3}, \end{aligned}$$

for some finite constant $\Gamma_{\text{OG}}(a, b) > 0$, where we recall that $\hat{f}_{1,a,b}^{\text{OG}} \equiv 1$ and that $\text{KL}(q \| 1)$ is finite. This completes the proof for the OG estimator.

Bounding the risk of the EA estimator. Fix $q \in \mathcal{D}_{a,b}$. Let

$$\mathcal{N}(\varepsilon, \mathcal{D}_{a,b}, d_{\text{H}}) \subset \mathcal{D}_{a,b}$$

be an ε -net of $\mathcal{D}_{a,b}$ with respect to the Hellinger distance d_{H} . Since $q \in \mathcal{D}_{a,b}$, there exists $q^* \in \mathcal{N}(\varepsilon, \mathcal{D}_{a,b}, d_{\text{H}})$ such that $d_{\text{H}}(q, q^*) \leq \varepsilon$. By slight abuse of notation, we write $q^* \equiv (q^*)_{t \geq 1}$. By definition,

$$\text{Risk}_{\mathbb{Q}}(q^*; n) = n \text{KL}(q \| q^*).$$

By Lemma 12, there exists a constant $C_1(a, b) > 0$ such that

$$\text{KL}(q \| q^*) \leq C_1(a, b) d_{\text{H}}^2(q, q^*) \leq C_1(a, b) \varepsilon^2.$$

Hence,

$$\text{Risk}_{\mathbb{Q}}(q^*; n) \leq C_1(a, b) n \varepsilon^2 \tag{16}$$

Next, we enumerate the elements of $\mathcal{N}(\varepsilon, \mathcal{D}_{a,b}, d_{\text{H}})$ as $\{g_1, \dots, g_{m_\varepsilon}\}$, and consider the exponentially weighted estimator over this finite class:

$$\hat{f}_{t,a,b}^{\text{EA}}(x) = \sum_{g \in \mathcal{N}(\varepsilon, \mathcal{D}_{a,b}, d_{\text{H}})} w_{t-1}(g) g(x), \quad w_{t-1}(g) \propto \frac{1}{m_\varepsilon} \prod_{s=1}^{t-1} g(X_s).$$

By Lemma 13, we have that

$$\sum_{s=1}^n \log \hat{f}_{s,a,b}^{\text{EA}}(X_s) \geq \max_{g \in \mathcal{N}(\varepsilon, \mathcal{D}_{a,b}, d_{\text{H}})} \sum_{s=1}^n \log g(X_s) - \log m_\varepsilon.$$

In particular, taking $g = q^*$ and expectation over $X_1, \dots, X_n \stackrel{iid}{\sim} \mathbb{Q}$, we obtain

$$\mathbb{E}_{\mathbb{Q}} \left[\sum_{s=1}^n \log q^*(X_s) - \sum_{s=1}^n \log \hat{f}_{s,a,b}^{\text{EA}}(X_s) \right] \leq \log m_{\varepsilon}. \quad (17)$$

Finally, by [Gao and Wellner \(2007, Theorem 1.1\)](#), the class of bounded monotone densities on $[0, 1]$ satisfies

$$\log N(\varepsilon, \mathcal{D}, \|\cdot\|_2) \leq C_2 \varepsilon^{-1}$$

for some constant $C_2 > 0$. Since $\mathcal{D}_{a,b} \subset \mathcal{D}$, the same bound holds for $\mathcal{D}_{a,b}$. Moreover, by [Lemma 12](#), the L_2 norm and the Hellinger distance are equivalent on $\mathcal{D}_{a,b}$, yielding

$$\log m_{\varepsilon} = \log N(\varepsilon, \mathcal{D}_{a,b}, d_H) \leq C_2 \varepsilon^{-1}.$$

Combining [\(16\)](#) and [\(17\)](#), we obtain

$$\text{Risk}_{\mathbb{Q}}(\hat{f}_{a,b}^{\text{EA}}; n) \leq C(a, b) \left(n\varepsilon^2 + \frac{1}{\varepsilon} \right),$$

for a constant $C(a, b) > 0$ depending only on a and b . Choosing $\varepsilon = n^{-1/3}$ yields the desired bound. ■

A.2. Completing the proof of [Theorem 3](#): pathwise regret bound

A.2.1. PROOF OF [LEMMA 5](#)

Fix $k \geq 1$ and set $V := \log(b/a)$, $\delta_k := a(e^{V/k} - 1)$ and for notational convenience, write $f_k^+ := \tilde{f}_{n,a+\delta_k,b}^{\text{MLE}}$. By [Lemma 10](#), f_k^+ admits a histogram representation

$$f_k^+(x) = \sum_{i=1}^{m_n} w_i \mathbb{1}_{\{t_{i-1} \leq x < t_i\}},$$

where $b \geq w_1 \geq \dots \geq w_{m_n} \geq a + \delta_k$, $0 = t_0 < t_1 < \dots < t_{m_n} = 1$ and $t_{m_n-1} \leq X_{(n)}$.

Now, in order to define the compressed histogram, we start by setting $i'_0 = 0$ and, for $\ell \geq 1$,

$$i'_\ell := \max \left\{ j \in \{i'_{\ell-1} + 1, \dots, m_n\} : \log w_{i'_{\ell-1}+1} - \log w_j \leq V/k \right\}.$$

If $i'_\ell = m_n$, the recursion terminates. Let L be the resulting number of blocks. Note that each time a new block begins, the log-height drops by at least V/k , while the total variation of $\log f_k^+$ is at most V . Hence $L \leq k$.

For each block $\ell = 1, \dots, L$, define the block interval

$$[t'_{\ell-1}, t'_\ell] := [t_{i'_{\ell-1}}, t_{i'_\ell}], \quad \text{so that} \quad 0 = t'_0 < t'_1 < \dots < t'_L = 1.$$

Since $i'_L = m_n$, we have $t'_{L-1} = t_{i'_{L-1}} \leq t_{m_n-1} \leq X_{(n)}$.

Now, for the compressed histogram, we assign to the bin $[t'_{\ell-1}, t'_\ell]$ a height proportional to the left-end height of each block, i.e.

$$\theta_\ell = \alpha w_{i'_{\ell-1}+1}, \quad \ell = 1, \dots, L,$$

for a suitable $\alpha > 0$, and we set

$$h(x) := \sum_{\ell=1}^L \theta_\ell \mathbb{1} \{t'_{\ell-1} \leq x < t'_\ell\}.$$

Observe that we can choose $\alpha > 0$ suitably, so that the $h(x)$ integrates to 1, i.e.,

$$\alpha \sum_{\ell=1}^L (t'_\ell - t'_{\ell-1}) w_{i'_{\ell-1}+1} = 1$$

Equivalently, define

$$\bar{h}(x) := \sum_{\ell=1}^L w_{i'_{\ell-1}+1} \mathbb{1} \{t'_{\ell-1} \leq x < t'_\ell\}, \quad h := \alpha \bar{h},$$

where $\alpha := \left(\int_0^1 \bar{h}(u) du \right)^{-1}$. Observe that $\ell \in \{1, \dots, L-1\}$, by the definition of $\{i'_\ell\}_{\ell \geq 1}$, we have $\log w_{i'_{\ell-1}+1} - \log w_{i'_\ell+1} > V/k$, and therefore

$$\log \theta_\ell - \log \theta_{\ell+1} = \log w_{i'_{\ell-1}+1} - \log w_{i'_\ell+1} \geq \frac{V}{k}.$$

This proves the first part.

For the second part, fix any $x \in [t'_{\ell-1}, t'_\ell)$ for some ℓ . Then, $\log w_{i'_{\ell-1}+1} - \log w_j \leq \frac{V}{k}$ implies that

$$\log(\bar{h}(x)) - \log f_k^+(x) \leq V/k.$$

Equivalently, we may write

$$\bar{h}(x) = w_{i'_{\ell-1}+1} \leq w_j e^{V/k} = f_k^+(x) e^{V/k}.$$

This holds for any $\ell \in \{1, \dots, L\}$ and any $x \in [t'_{\ell-1}, t'_\ell)$. By construction, for every $x \in [0, 1]$,

$$f_k^+(x) \leq \bar{h}(x) \leq e^{V/k} f_k^+(x).$$

Therefore, $1 \leq \int_0^1 \bar{h}(u) du \leq e^{V/k}$, and hence $e^{-V/k} \leq \alpha \leq 1$. Since $\bar{h}(x) \geq a + \delta_k$ and $\bar{h}(x) \leq b$, we obtain

$$h(x) \geq e^{-V/k} (a + \delta_k) = a, \quad h(x) \leq \bar{h}(x) \leq b.$$

Thus $h \in \mathcal{D}_{a,b}$. Moreover, for all $x \in [0, 1]$,

$$|\log f_k^+(x) - \log h(x)| = |\log f_k^+(x) - \log(\bar{h}(x)) + \log(\alpha)| \leq \frac{2V}{k}.$$

Finally, for the observed sample path $(x_1, \dots, x_n)$,

$$|\mathcal{L}(f_k^+, n) - \mathcal{L}(h, n)| = \left| \sum_{t=1}^n \log f_k^+(x_t) - \log h(x_t) \right| \leq \frac{2nV}{k}.$$

This completes the proof. ■

A.2.2. PROOF OF LEMMA 6

Proof of first part: Given a f of the form

$$f(x) = \sum_{j=1}^r \theta_j \mathbb{1}_{\{t_{j-1} \leq x < t_j\}}, \quad 0 = t_0 < t_1 < \dots < t_r = 1, \quad r \leq k,$$

as in the theorem hypothesis, we construct $g \in \mathcal{E}_{k,n,\beta}$ as follows whenever (10) holds:

Endpoint approximation: First, for each $j = 1, \dots, r-1$, define the left-rounded endpoints

$$t'_j := \max\{u \in \mathcal{B}_n : u \leq t_j\}.$$

Then $t_j - \Delta_b \leq t'_j \leq t_j$. Set $t'_0 = 0$ and $t'_r = 1$.

Log-height approximation: Next, for $j = 1, \dots, r-1$, define the down-rounded log-heights

$$\ell'_j := \max\{\ell \in \Lambda_{n,a,b} : \ell \leq \log \theta_j\}, \quad \theta'_j := e^{\ell'_j}.$$

We write $\eta := V\Delta_b$. By construction, $\log \theta_j - \eta \leq \ell'_j \leq \log \theta_j$. By (10), $\eta < 1$ and

$$\theta_j \geq \theta'_j \geq e^{-\eta} \theta_j \quad \text{and} \quad \theta_j - \theta'_j \leq (1 - e^{-\eta}) \theta_j \leq b\eta. \quad (18)$$

Moreover, since the map $\ell \mapsto \max\{\ell' \in \Lambda_{n,a,b} : \ell' \leq \ell\}$ is monotone,

$$b \geq \theta'_1 \geq \dots \geq \theta'_{r-1} \geq a.$$

Final height assignment: Finally, we define θ'_r by normalization:

$$\theta'_r := \frac{1 - \sum_{j=1}^{r-1} \theta'_j (t'_j - t'_{j-1})}{1 - t'_{r-1}}.$$

Consequently, we define the histogram

$$g(x) := \sum_{j=1}^r \theta'_j \mathbb{1}_{\{t'_{j-1} \leq x < t'_j\}}.$$

In order to show that $g \in \mathcal{E}_{k,n,\beta}$ we need to show that $\theta'_r \geq a$ and $\theta'_{r-1} \geq \theta'_r$.

We first show that $\theta'_r \geq \theta_r \geq a$. Suppose for a contradiction that $\theta'_r < \theta_r$. Since $t'_j \leq t_j$ and $\theta'_j \leq \theta_j$ for all $j \leq r-1$, the left-rounded partition and down-rounded heights imply the pointwise domination $g(x) \leq f(x)$ for all $x \in [0, 1]$. In particular,

$$\sum_{j=1}^r \theta'_j (t'_j - t'_{j-1}) = \int_0^1 g(x) dx < \int_0^1 f(x) dx = 1,$$

This is a contradiction, since $\int_0^1 g(x) dx = 1$ by definition. This proves that $\theta'_r \geq a$.

To prove that $\theta'_{r-1} \geq \theta'_r$, we start by upper bounding θ'_r . Using normalization for f and g , we may write

$$\theta'_r = \frac{\theta_r(1 - t_{r-1})}{1 - t'_{r-1}} + \frac{\sum_{j=1}^{r-1} \theta_j(t_j - t_{j-1}) - \sum_{j=1}^{r-1} \theta'_j(t'_j - t'_{j-1})}{1 - t'_{r-1}}.$$

Since $t'_{r-1} \leq t_{r-1}$, the first term is at most θ_r . For the second term, we bound the numerator by

$$\begin{aligned} & \sum_{j=1}^{r-1} \theta_j(t_j - t_{j-1}) - \sum_{j=1}^{r-1} \theta'_j(t'_j - t'_{j-1}) \\ & \leq \sum_{j=1}^{r-1} (\theta_j - \theta'_j)(t_j - t_{j-1}) + \sum_{j=1}^{r-1} \theta'_j(|t'_j - t_j| + |t'_{j-1} - t_{j-1}|) \\ & \leq b\eta + 2b(r-1)\Delta_b \leq b(\eta + 2r\Delta_b), \end{aligned}$$

where we used $|t'_j - t_j| \leq \Delta_b$ and $|t'_{j-1} - t_{j-1}| \leq \Delta_b$. Finally, since $1 - t_{r-1} \geq n^{-\gamma}$ and $t'_{r-1} \leq t_{r-1}$, we have $1 - t'_{r-1} \geq n^{-\gamma}$, and therefore

$$\theta'_r \leq \theta_r + b(\eta + 2r\Delta_b)n^\gamma \leq \theta_r + b(\eta + 2k\Delta_b)n^\gamma.$$

By the theorem hypothesis, by (10), $V/k \leq 1$, and

$$\theta_{r-1} - \theta_r = \theta_r(e^{\log \theta_{r-1} - \log \theta_r} - 1) \geq a(e^{V/k} - 1) \geq a \frac{V}{k}.$$

By (18), $\theta'_{r-1} \geq \theta_{r-1} - b\eta$ and $\theta'_r \leq \theta_r + b(\eta + 2k\Delta_b)n^\gamma$, the condition (10) guarantees $\theta'_r \leq \theta'_{r-1}$, and hence $g \in \mathcal{E}_{k,n,\beta}$. This proves the feasibility of g , and completes the first part.

Proof of second part: Now, for the second part, we start by defining the boundary neighborhoods

$$I_j := (t'_j, t_j], \quad j = 1, \dots, r-1.$$

Each I_j has length at most $\Delta_b < n^{-\beta}$, and since the sample path belongs in $\mathcal{S}_n(\beta, \gamma)$, defined in (7), I_j may contain at most one sample point. Hence at most $r-1$ indices $t \in \{1, \dots, n\}$ can change bin membership when the partition changes from $(t_j)_{j=1,\dots,r}$ to $(t'_j)_{j=1,\dots,r}$. In particular, we partition indices $i \in \{1, \dots, n\}$ into two types.

- If $x_i \notin \bigcup_{j=1}^{r-1} I_j$, then x_i falls in the same bin under f and g , say bin ℓ , and

$$|\log f(x_i) - \log g(x_i)| = |\log \theta_\ell - \log \theta'_\ell| \leq \eta.$$

- If $x_i \in \bigcup_{j=1}^{r-1} I_j$, then bin membership may change. There are at most $r-1$ such indices, and for these we use the bound

$$|\log f(x_i) - \log g(x_i)| \leq \log(b/a) = V.$$

Combining the two cases yields

$$\mathcal{L}(g, n) - \mathcal{L}(f, n) \leq n\eta + (r-1)V \leq n\eta + (k-1)V.$$

as claimed. ■

A.3. Proof of results from Section 3

A.3.1. PROOF OF THEOREM 8: ASYMPTOTIC LOG-OPTIMALITY OF CALIBRATORS

We start with recalling that, since $q \in \mathcal{D}$, the log-optimal decreasing calibrator satisfies

$$h_t^{\text{opt}} = q \quad \text{for all } t \in \mathbb{N}.$$

Consequently, we have

$$\log M_n(h^{\text{opt}}) = \sum_{s=1}^n \log q(p_s), \quad \log M_n(\hat{h}) = \sum_{s=1}^n \log \hat{h}_s(p_s),$$

for any online calibrator $\hat{h} = (\hat{h}_s)$. Now, we write

$$\log M_n(\hat{h}) - \log M_n(h^{\text{opt}}) = \sum_{t=1}^n \log \frac{\hat{h}_t(P_t)}{q(P_t)} =: \sum_{t=1}^n \Delta_t,$$

and define

$$a_t := \mathbb{E}_{\mathbb{Q}}[\Delta_t \mid \mathcal{F}_{t-1}] = \mathbb{E}_{\mathbb{Q}} \left[\log \frac{\hat{h}_t(P_t)}{q(P_t)} \mid \mathcal{F}_{t-1} \right] = -\text{KL}(q \parallel \hat{h}_t).$$

We decompose

$$\sum_{t=1}^n \Delta_t = S_n - A_n, \quad S_n := \sum_{t=1}^n (\Delta_t - a_t), \quad A_n := -\sum_{t=1}^n a_t = \sum_{t=1}^n \text{KL}(q \parallel \hat{h}_t).$$

By construction, $\{S_n\}_{n \geq 0}$ is a martingale with respect to $\{\mathcal{F}_n\}$. Since $q, \hat{h}_t \in \mathcal{D}_{a,b}$ by the theorem hypothesis, we have

$$|\Delta_t| \leq \log(b/a), \quad |\Delta_t - a_t| \leq 2 \log(b/a).$$

Therefore, by the Azuma–Hoeffding inequality, for any $\varepsilon > 0$,

$$\mathbb{P}_{\mathbb{Q}}(|S_n| \geq n\varepsilon) \leq 2 \exp \left(-\frac{n\varepsilon^2}{8(\log(b/a))^2} \right).$$

Since $\sum_{n=1}^{\infty} e^{-cn} < \infty$ for any $c > 0$, the first Borel–Cantelli lemma implies

$$\frac{S_n}{n} \rightarrow 0 \quad \mathbb{Q}\text{-almost surely.}$$

It remains to show that $A_n/n \rightarrow 0$ almost surely. By Theorem 2, for $\hat{h} \in \{\hat{f}_{a,b}^{\text{OG}}, \hat{f}_{a,b}^{\text{EA}}\}$, there exists $C = C(a, b) > 0$ such that

$$\mathbb{E}_{\mathbb{Q}}[A_n] \leq C n^{1/3}.$$

Consider the dyadic subsequence $n_k = 2^k$. By Markov’s inequality, for any $\epsilon > 0$,

$$\mathbb{P}_{\mathbb{Q}} \left(\frac{A_{n_k}}{n_k} \geq \epsilon \right) \leq \frac{C}{\epsilon} 2^{-2k/3}.$$

Since $\sum_{k \geq 1} 2^{-2k/3} < \infty$, the first Borel–Cantelli lemma yields

$$\frac{1}{2^k} A_{2^k} \rightarrow 0 \quad \mathbb{Q}\text{-almost surely.}$$

Finally, since $\sum_{t=1}^n \text{KL}(q \parallel \hat{h}_t)$ is non-decreasing in n , for $2^k \leq n < 2^{k+1}$,

$$\frac{A_n}{n} \leq 2 \cdot \frac{A_{2^{k+1}}}{2^{k+1}}$$

which converges to zero almost surely. Combining the two parts proves

$$\frac{1}{n} (\log M_n(\hat{h}) - \log M_n(h^{\text{opt}})) \rightarrow 0 \quad \mathbb{Q}\text{-almost surely.}$$

■

A.3.2. PROOF OF COROLLARY 9

Since $h_t^{\text{opt}} = q$ for all $t \in \mathbb{N}$, by the strong law of large numbers, we have

$$\frac{1}{n} \log M_n(h^{\text{opt}}) = \frac{1}{n} \sum_{s=1}^n \log q(p_s) \xrightarrow{\text{a.s.}} \mathbb{E}_{\mathbb{Q}}[\log q] = \int_0^1 q(u) \log q(u) du.$$

By Theorem 8, the proof follows. ■

Appendix B. Supporting lemmas

Lemma 10 (Histogram form of the constrained Grenander) *Fix $n \in \mathbb{N}$ and let $0 = x_{(0)} < x_{(1)} < \dots < x_{(r_n)} < x_{(r_n+1)} = 1$ denote the distinct order statistics of the sample path $x_1, \dots, x_n$, where $r_n \leq n$. Further, fix $0 < a \leq b < \infty$, and let $\tilde{f}_{n,a,b}^{\text{MLE}}$ denote the constrained Grenander estimator, defined in (3).*

Then $\tilde{f}_{n,a,b}^{\text{MLE}}$ can be chosen to be a monotone histogram of the form

$$\tilde{f}_{n,a,b}^{\text{MLE}}(u) = \theta_j, \quad u \in (x_{(j)}, x_{(j+1)}], \quad j = 0, \dots, r_n,$$

where

$$b \geq \theta_0 \geq \theta_1 \geq \dots \geq \theta_{r_n} \geq a.$$

In particular, the knots of the MLE may be taken at the order statistics.

Proof Let $f \in \mathcal{D}_{a,b}$ be any candidate density, and define its cumulative distribution function

$$F(x) := \int_0^x f(u) du.$$

Since f is non-increasing, F is concave and absolutely continuous on $[0, 1]$, with

$$a \leq F'(x) = f(x) \leq b \quad \text{a.e.}$$

Next, define a piecewise-linear function $\tilde{F}$ by chord interpolation of F at the knots $\{x_{(j)}\}_{j=0}^{r_n+1}$:

$$\tilde{F}(u) := F(x_{(j)}) + \theta_j (u - x_{(j)}), \quad u \in [x_{(j)}, x_{(j+1)}],$$

where we define

$$\theta_j := \frac{F(x_{(j+1)}) - F(x_{(j)})}{x_{(j+1)} - x_{(j)}}.$$

Since F is concave, the slopes must satisfy

$$\theta_0 \geq \theta_1 \geq \cdots \geq \theta_{r_n}.$$

Moreover, since

$$\theta_j = \frac{1}{x_{(j+1)} - x_{(j)}} \int_{x_{(j)}}^{x_{(j+1)}} f(u) du \in [a, b], \quad (19)$$

$\tilde{F}$ is a concave CDF whose derivative $\tilde{f} := \tilde{F}'$ belongs to $\mathcal{D}_{a,b}$. At each distinct observation $x_{(j)}$, by (19),

$$\tilde{f}(x_{(j)}) = \theta_{j-1} \geq f(x_{(j)}).$$

Hence, it follows that

$$\sum_{i=1}^n \log f(x_i) \leq \sum_{i=1}^n \log \tilde{f}(x_i).$$

Applying this argument with $f = \tilde{f}_{n,a,b}^{\text{MLE}}$ shows that any maximizer of the likelihood may be replaced by its chord-interpolated version without decreasing the likelihood. Therefore, $\tilde{f}_{n,a,b}^{\text{MLE}}$ can be chosen to be piecewise constant with knots at the order statistics, as claimed. $\blacksquare$

Lemma 11 *Let $X_1, \dots, X_n \stackrel{iid}{\sim} \mathbb{Q}$, where $\mathbb{Q}$ is supported on $[0, 1]$ and has density $q \in \mathcal{D}_{a,b}$. Fix $\beta > 2$ and $\gamma > 1$. Then, there exists $N \equiv N(\delta, \beta, \gamma, b)$ such that for all $n > N$,*

$$\mathbb{P}_{\mathbb{Q}} \left(\min_{1 \leq i < j \leq n} |X_i - X_j| \geq n^{-\beta}, \max_{1 \leq i \leq n} X_i \leq 1 - n^{-\gamma} \right) \geq 1 - \delta.$$

Proof Let $\tau = n^{-\beta}$. By a union bound,

$$\mathbb{P}_{\mathbb{Q}} \left(\min_{1 \leq i < j \leq n} |X_i - X_j| \leq \tau \right) \leq \binom{n}{2} \mathbb{P}_{X'_1, X'_2 \stackrel{iid}{\sim} \mathbb{Q}} (|X'_1 - X'_2| \leq \tau).$$

Since $\mathbb{Q}$ has density bounded above by b ,

$$\mathbb{P}_{X'_1, X'_2 \stackrel{iid}{\sim} \mathbb{Q}} (|X'_1 - X'_2| \leq \tau \mid X_1) \leq \frac{2bn^2}{2n^\beta} = \frac{b}{n^{\beta-2}}.$$

Next, we note that for any distribution $\mathbb{Q}$ with density $q \in \mathcal{D}_{a,b}$, $\mathbb{Q}$ is stochastically dominated by $\text{Unif}[0, 1]$. Therefore,

$$\mathbb{P}_{\mathbb{Q}} \left(\max_{1 \leq i \leq n} X_i \geq 1 - n^{-\gamma} \right) \leq \mathbb{P}_{\text{Unif}[0,1]} \left(\max_{1 \leq i \leq n} X_i \geq 1 - n^{-\gamma} \right)$$

Now, by a union bound,

$$\mathbb{P}_{\text{Unif}[0,1]} \left(\max_{1 \leq i \leq n} X_i \geq 1 - n^{-\gamma} \right) \leq n^{-\gamma+1}$$

Combining both probability bounds by a union bound, the existence of a $N \equiv N(\delta, \beta, \gamma)$ is immediate, such that for all $n > N$,

$$\mathbb{P}_{\mathbb{Q}}\left(\min_{1 \leq i < j \leq n} |X_i - X_j| \geq n^{-\beta}, \max_{1 \leq i \leq n} X_i \leq 1 - n^{-\gamma}\right) \geq 1 - \delta.$$

This completes the proof. ■

Lemma 12 *Let $f, g \in \mathcal{D}_{\gamma_0, \gamma_1}$, where $0 < \gamma_0 \leq \gamma_1 < \infty$. Then the following bounds hold:*

1. $\text{KL}(f\|g) \leq \frac{1}{\gamma_0} \int_0^1 (f(u) - g(u))^2 du.$
2. $\text{KL}(f\|g) \leq \frac{8\gamma_1}{\gamma_0} d_{\text{H}}^2(f, g)$
3. $\|\cdot\|_2$ and d_{H} are equivalent on $\mathcal{D}_{\gamma_0, \gamma_1}$ in the sense that

$$\gamma_0 \int_0^1 (\sqrt{f(u)} - \sqrt{g(u)})^2 du \leq \int_0^1 (f(u) - g(u))^2 du \leq 4\gamma_1 \int_0^1 (\sqrt{f(u)} - \sqrt{g(u)})^2 du.$$

Proof We prove each part separately.

Proof of part 1. We write

$$\text{KL}(f\|g) = \int_0^1 f(x) \log \frac{f(x)}{g(x)} dx = \int_0^1 f(x) \log \left(1 + \frac{f(x) - g(x)}{g(x)}\right) dx.$$

Since $f(x) \geq \gamma_0$, we have $\frac{f(x)-g(x)}{g(x)} > -1$, and hence $\log(1+u) \leq u$ yields

$$\text{KL}(f\|g) \leq \int_0^1 f(x) \frac{f(x) - g(x)}{g(x)} dx = \int_0^1 \frac{(f(x) - g(x))^2}{g(x)} dx + \int_0^1 (f(x) - g(x)) dx.$$

The last integral is zero because f and g are in $\mathcal{D}$, so

$$\text{KL}(f\|g) \leq \int_0^1 \frac{(f(x) - g(x))^2}{g(x)} dx \leq \frac{1}{\gamma_0} \int_0^1 (f(x) - g(x))^2 dx,$$

which proves the first bound.

Proof of part 2. For the second bound, note that

$$(f(x) - g(x))^2 = (\sqrt{f(x)} - \sqrt{g(x)})^2 (\sqrt{f(x)} + \sqrt{g(x)})^2.$$

Since $f(x), g(x) \leq \gamma_1$, we have $(\sqrt{f(x)} + \sqrt{g(x)})^2 \leq 4\gamma_1$, and thus

$$\int_0^1 (f(x) - g(x))^2 dx \leq 4\gamma_1 \int_0^1 (\sqrt{f(x)} - \sqrt{g(x)})^2 dx.$$

Combining this with the first bound and $d_{\text{H}}^2(f, g) = \frac{1}{2} \int_0^1 (\sqrt{f(x)} - \sqrt{g(x)})^2 dx$ gives

$$\text{KL}(f\|g) \leq \frac{1}{\gamma_0} \int_0^1 (f(x) - g(x))^2 dx \leq \frac{4\gamma_1}{\gamma_0} \int_0^1 (\sqrt{f(x)} - \sqrt{g(x)})^2 dx = \frac{8\gamma_1}{\gamma_0} d_{\text{H}}^2(f, g).$$

Proof of part 3. Finally, using again $(f(x)-g(x))^2 = (\sqrt{f(x)}-\sqrt{g(x)})^2(\sqrt{f(x)}+\sqrt{g(x)})^2$ and the bounds

$$(\sqrt{f(x)} + \sqrt{g(x)})^2 \leq 4\gamma_1, \quad (\sqrt{f(x)} - \sqrt{g(x)})^2 \geq (\sqrt{f(x)})^2 \geq \gamma_0,$$

we obtain

$$\gamma_0 \int_0^1 (\sqrt{f(x)} - \sqrt{g(x)})^2 dx \leq \int_0^1 (f(x) - g(x))^2 dx \leq 4\gamma_1 \int_0^1 (\sqrt{f(x)} - \sqrt{g(x)})^2 dx,$$

which implies the displayed equivalence. $\blacksquare$

Lemma 13 *Let $g_1, \dots, g_M \in \mathcal{D}$, and let $\pi_1, \dots, \pi_M$ be prior weights with $\pi_j > 0$ for all j and $\sum_{j=1}^M \pi_j = 1$. Given observations $x_1, \dots, x_n$, define for each $t \in [n]$ the weighted mixture*

$$\hat{f}_t(x) = \sum_{j=1}^M w_{t-1}(j) g_j(x), \quad w_{t-1}(j) = \frac{\pi_j \prod_{s=1}^{t-1} g_j(x_s)}{\sum_{\ell=1}^M \pi_\ell \prod_{s=1}^{t-1} g_\ell(x_s)}.$$

Then, for every $j \in \{1, \dots, M\}$,

$$\sum_{t=1}^n \log \hat{f}_t(x_t) \geq \sum_{t=1}^n \log g_j(x_t) + \log \pi_j.$$

In particular, if $\pi_j = 1/M$ for all $j \in [M]$, then

$$\sum_{t=1}^n \log \hat{f}_t(x_t) \geq \max_{1 \leq j \leq M} \sum_{t=1}^n \log g_j(x_t) - \log M.$$

Proof The result is standard; see, for instance, [Cesa-Bianchi and Lugosi \(2006, Section 9.2\)](#). We include the argument here for completeness.

Define the normalizing constants

$$Z_t := \sum_{\ell=1}^M \pi_\ell \prod_{s=1}^t g_\ell(x_s), \quad Z_0 := \sum_{\ell=1}^M \pi_\ell = 1.$$

By the definition of w_{t-1} , for each $t \in [n]$,

$$\hat{f}_t(x_t) = \sum_{j=1}^M w_{t-1}(j) g_j(x_t) = \frac{\sum_{j=1}^M \pi_j \prod_{s=1}^{t-1} g_j(x_s) g_j(x_t)}{\sum_{\ell=1}^M \pi_\ell \prod_{s=1}^{t-1} g_\ell(x_s)} = \frac{Z_t}{Z_{t-1}}.$$

Therefore,

$$\sum_{t=1}^n \log \hat{f}_t(x_t) = \sum_{t=1}^n (\log Z_t - \log Z_{t-1}) = \log Z_n - \log Z_0 = \log Z_n.$$

Since $Z_n = \sum_{\ell=1}^M \pi_\ell \prod_{s=1}^n g_\ell(x_s) \geq \pi_j \prod_{s=1}^n g_j(x_s)$, we obtain

$$\log Z_n \geq \log \pi_j + \sum_{t=1}^n \log g_j(x_t),$$

which proves the first claim. The second follows by taking $\pi_j = 1/M$ and maximizing over j . $\blacksquare$